Beamer Live Preview

A demo deck

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Basics

Lists and emphasis

- Plain text with **bold**, *emphasis*, alerts and `monospace`.
- Nested lists:
  - second level is `\small`
  - and uses the same bullet
- Enumerations:
  1. first
  2. second

Overlays

- Revealed step by step

Overlays

- Revealed step by step
- with `\pause`

Overlays

- Revealed step by step
- with `\pause`
- and the preview shows step dots in the corner

Step three replaces it again.

Math

- Inline $E(X) = \sum_i p(x_i)\, x_i$ and a user macro $\|x\|_2$.
- Display math:

$$\hat{\beta} = \left(X^\top X\right)^{-1} X^\top y$$

- Aligned equations:

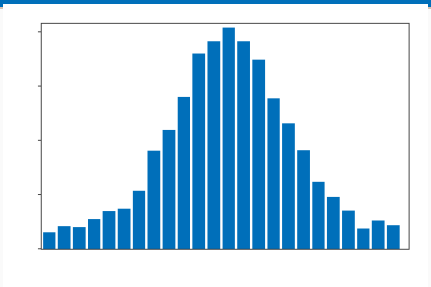
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

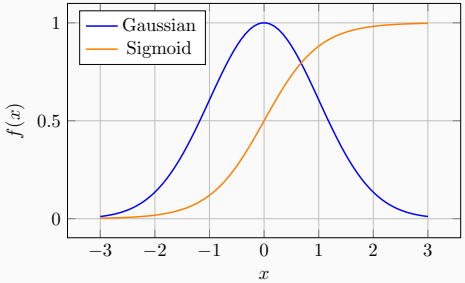
$$\sigma(x) = \frac{1}{1+e^{-x}}$$

Columns and images

- Left: a PNG image
- Right: a PDF figure, rendered with pdf.js
- Sizes follow `width=` / `height=`[a]

[a] Footnotes go to the bottom of the frame.





Blocks

Block

A regular block with `block=fill`.

Alert block

Titles use the alert colour.

Example block

Titles use the `example text` colour.

Advanced

Tables

Model	Accuracy	F1
Logistic regression	0.91	0.88
Random forest	0.94	0.92
Gradient boosting	0.95	0.93

Table 1: Booktabs rules and a coloured row.

	Predicted +	Predicted −
Actual +	TP	FN
Actual −	FP	TN

TikZ



- `tikzpicture` is compiled once in the background and cached.

Algorithms

Algorithm 1: K-Means(Dataset: D, Clusters: k)

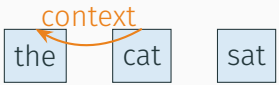
```
1 begin
2     Choose k initial centroids;
3     repeat
4         Assign each point to its closest centroid;
5         Recompute each centroid as the mean of its points;
6     until centroids do not change;
7     return clusters;
8 end
```

TikZ with overlays



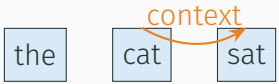
- `\visible<n>` inside a picture: one compiled variant per overlay step.

TikZ with overlays



- `\visible<n>` inside a picture: one compiled variant per overlay step.

TikZ with overlays



- `\visible<n>` inside a picture: one compiled variant per overlay step.

Listings

```
import numpy as np  # keywords, comments and strings are highlighted

def softmax(z):
    e = np.exp(z - z.max())
    return e / e.sum()   # "numerically stable"
```

Further reading

[1] The beamer class

[2] The metropolis theme

[3] KaTeX

Bibliography i

1. Hochreiter, Sepp, and Jürgen Schmidhuber. “Long short-term memory.” Neural Computation 9, no. 8 (1997): 1735–1780.
2. LeCun, Yann, Léon Bottou, Yoshua Bengio, and Patrick Haffner. “Gradient-based learning applied to document recognition.” Proceedings of the IEEE 86, no. 11 (1998): 2278–2324.
3. Krizhevsky, Alex, Ilya Sutskever, and Geoffrey E. Hinton. “ImageNet classification with deep convolutional neural networks.” Advances in Neural Information Processing Systems 25 (2012).
4. Kingma, Diederik P., and Max Welling. “Auto-encoding variational Bayes.” International Conference on Learning Representations (2014).

Bibliography ii

5. Goodfellow, Ian, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. “Generative adversarial nets.” Advances in Neural Information Processing Systems 27 (2014).
6. He, Kaiming, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. “Deep residual learning for image recognition.” In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 770–778. 2016.
7. Vaswani, Ashish, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. “Attention is all you need.” Advances in Neural Information Processing Systems 30 (2017).

Code

Verbatim

```
from sklearn.cluster import KMeans
model = KMeans(n_clusters=3).fit(X)
print(model.cluster_centers_)
```

Questions?